FAHEEM NAZIR

Software Engineer (Fresher) • Full Stack • Backend • AI/ML

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PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student with hands-on experience building full-stack applications, backend APIs and AI-powered features. Built a multithreaded C++17 search engine on SQLite, a FastAPI and React research assistant with PDF processing and source-grounded RAG, and a role-based placement platform (Node.js, MongoDB) with Gemini integration. Solid grounding in data structures, databases and automated testing.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript
Frontend: React.js, HTML, CSS, Tailwind CSS
Backend: FastAPI, Node.js, Express.js, REST APIs, JWT authentication, Firebase Authentication
Databases: SQLite, MongoDB (Mongoose), MySQL, PostgreSQL
AI / ML: NLP, Generative AI, Retrieval-Augmented Generation, Sentence Transformers, PyMuPDF, scikit-learn, TensorFlow, NumPy, Pandas
Tools: Git, GitHub, Linux, Docker, Postman, CMake, CTest
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

FLUX | *Native Windows File Search Engine*

Tech: *C++17, SQLite, CMake, CTest, multithreading*

- Engineered a fast, offline file search engine for Windows with a native Win32 GUI and a CLI that share one service layer, backed by a SQLite inverted index.
- Designed a multithreaded indexing pipeline (bounded producer-consumer queue, worker pool, dedicated database writer thread) with pause, resume, cancellation and race-safe file handling.
- Implemented transactional incremental sync with change detection, plus schema migration with backup and rollback.
- Built a benchmark harness with correctness checks: ~3.7M tokens/sec at 8 threads (~3.5x speedup on the scan, read and tokenize stage); ~1.3x with database writes enabled, limited by the single-writer design.
- Verified with CMake/CTest: 23 test suites, 99 GUI smoke checks and 53 package checks, all passing at last recorded run. In progress: in-house text extractors for PDF, DOCX, XLSX and HTML (98.1% word recall vs. pdftotext on 53 PDFs in a prototype; not yet integrated).

Paper Pilot | *AI-Powered Scientific Literature Review Assistant*

Live | GitHub

Tech: *Python, FastAPI, PyMuPDF, SQLite, React 19, TypeScript, Firebase Auth, Sentence Transformers*

- Developed a full-stack app that turns uploaded scientific PDFs into structured, searchable documents, with Firebase-authenticated, per-user paper libraries.
- Built a PyMuPDF processing pipeline: text extraction, dehyphenation, ligature normalization, header/footer removal, and heuristic section detection (abstract, methods, results, etc.).
- Implemented document chunking and embedding-based retrieval for a RAG Research Mentor (Gemini) that returns source-cited answers and declines when evidence is insufficient.
- Evaluated an experimental LoRA fine-tuned Qwen2.5-0.5B model, found grounding failures, and moved to a more reliable extraction-based approach; backend covered by 302 tests.

HireSense AI | *AI-Powered Placement Preparation & Recruitment Platform*

Live | GitHub

Tech: *React, TypeScript, Node.js, Express.js, MongoDB, Mongoose, JWT, Zod, Zustand*

- Built a multi-role campus placement platform for Students, TPOs, Recruiters and Admins with role-specific dashboards, job drives, applications and eligibility management.
- Implemented JWT authentication, role-based authorization and OTP registration, with bcrypt password hashing and Zod request validation on a layered controller-service-model REST API.
- Integrated Google Gemini for resume analysis on uploaded, versioned PDF resumes (LLM-generated feedback and quality score); added company prep resources, previous-year questions and quizzes.

EDUCATION

B.Tech, Computer Science and Engineering | Kolhapur Institute of Technology | CGPA: 7.6 / 10 — 2023 - 2027
12th Grade | SRM Welkin Higher Secondary School, Sopore — 83%
10th Grade | SRM Welkin Higher Secondary School, Sopore — 98%

ACHIEVEMENTS & CERTIFICATIONS

- Solved 200+ Data Structures & Algorithms problems.
- Getting Started with Artificial Intelligence, IBM SkillsBuild.

Languages: English (Professional), Urdu/Hindi (Native), Kashmiri (Native)